Module 4: A+ Troubleshooting and Helpdesk

DAY 2:-

Task 1:

Hardware Issues	Symptoms
1. No display on monitor	Blank screen, no POST beep, power LED on but nothing shows.
2. RAM failure/poor seating	System beeps in a repeating pattern, fails to boot pass POST.
3. Failing/dying hard drive	Frequent freezing, “Disk Boot Failure” message , clicking noises (HDD).
4. Overheating CPU/GPU	Random shutdowns or restarts under load, thermal throttling, loud fan noise before crash.
5. Faulty power supply	PC won’t power on at all, random reboots, burning smell or intermittent shutdowns.

Task 2:

Troubleshooting Plan: Laptop Not Charging

Step 1: Check the power outlet
Confirm the wall outlet itself has power by plugging in another device to rule out a dead outlet before suspecting the laptop.

Step 2: Inspect the charger cable and adapter
Look at the charging cable and adapter for any visible damage.

Step 3: Reseat the charging connector
Unplug the charger from the laptop, then plug it back in firmly. A loose connection at the laptop’s charging port is one of the most common causes of intermittent or failed charging.

Step 4: Check the charging port for debris or damage
Using a flashlight , check inside the laptop’s charging port for dust, lint or bent pins that could be blocking a proper connection.

Step 5: Test with a different charger if available
If you have access to another compatible charger, try it on the laptop. If the laptop charges with a different charger, the original charger is likely faulty.

Step 6: Try a battery reset (if removable battery)
Fully power off the laptop, remove the battery if it's removable , wait 30s , then reinsert and try charging again.

Step 7: Determine if professional repair is needed
If none of the able works, the issue likely lies with the laptop's internal charging circuitry or a degraded battery.

Task 4:

Steps:

1. Power off the PC and disconnect the 24-pin ATX connector from the motherboard (but keep the PSU plugged into the wall and switched on)
2. Set the multimeter to DC voltage mode
3. Insert the black probe into any black (ground) wire pin on the connector
4. Insert the red probe into a yellow wire pin (this carries the +12V rail)
5. Power on the PSU (some require briefly jumping the green wire to a black/ground pin to force it on without a motherboard connected) and read the voltage
6. Record the exact value shown
7. If the voltage is below 11.5V then it indicates to a failing/degrading of the PSU.